Shuhan Wang

MSc Machine Learning | World Models / Model-Based RL

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PROFILE

UCL MSc Machine Learning candidate interested in how world models learned from an agent's existing experience can generate open-ended curricula that keep extending its abilities. Currently a Research Assistant at CityU Hong Kong, investigating where and why learned world models fail.

EDUCATION

University College London (UCL) — MSc Machine Learning Sep 2025 – Dec 2026 (expected)
Distinction (expected); taught-component average 79.3/100

University of Manchester — BSc Mathematics and Physics Sep 2022 – Jun 2025
First-Class Honours; overall average 76.8/100

RESEARCH EXPERIENCE

Research Assistant — City University of Hong Kong Oct 2026 – present
Supervisor: Prof. Yangchen Pan; independent project on diagnosing errors in learned world models

- **Diagnostic environment:** Building a controlled offline environment with known ground truth to locate world-model errors and test competing explanations of their causes.
- **Evaluation design:** Designing evaluations that vary what the model observes while holding environment dynamics fixed, with controlled faults and simple baselines to assess diagnostic reliability.

CLIMB: DreamerV3 for Open-Ended Continual Learning Feb 2026 – Apr 2026
UCL group research; project lead and first author of the course report

- **World-model-guided policy learning:** Proposed and implemented a soft feasibility mask for DreamerV3 in JAX / Flax on **Craftax-Symbolic**: an RSSM-based head predicted action context, and hand-specified feasibility rules used the environment's context during interaction and the head's predictions in imagined rollouts.
- **Exploration and continual learning:** Proposed and implemented episodic spatial and inventory novelty rewards and novelty–learnability–uniform replay, and ran component ablations to examine their contributions to exploration and achievement progression.
- **Controlled evaluation:** Evaluated 19 configurations across three seeds. At 10M steps, mean achievement rate over the final 200 episodes was 0.200 ± 0.005 versus 0.099 ± 0.008 for the matched 50:50 recent/uniform replay baseline (mean $\pm$ SD); ablations identified the soft feasibility mask as the main isolated contributor.

OpenQFT-Bench: Towards Open-Ended Learning Jul 2026 – present
Independent research; in development

- **Benchmark design:** Developing a benchmark of simplified particle-physics worlds with different physical laws, in which agents choose collision experiments, infer hidden particles and interaction rules, and revise world models from partial, noisy observations. The long-term objective is to test whether agents learn transferable discovery strategies.
- **Benchmark implementation:** Built the core loop in JAX: a simulator that generates hidden worlds, an environment in which agents run experiments, a Bayesian learner that updates its beliefs with Sequential Monte Carlo and picks the most informative next experiment, and checks of both its predictions and the rules it recovers.

RopeBand: RoPE-Compatible Q/K Rotations May 2026 – present
First author; under review at TMLR

- **Large-scale GPU experiments:** Ran multi-seed dynamic 4-bit quantisation experiments on four 1B–8B Llama / Mistral checkpoints, up to 128K context, as resumable SGE jobs on UCL's Myriad cluster; found higher perplexity than full-head Hadamard for the evaluated pairwise-only transform.

SWE-Router: Data and Evaluation for Agent Routing Apr 2026 – Jul 2026
Co-author; ICML 2026 DL4C Workshop

- **Benchmark evaluation and data collection:** Independently ran official SWE-bench / SWE-Smith grading in Docker / Apptainer containers and collected multi-model mini-SWE-agent trajectories with per-step cost annotation, producing public Hugging Face trajectory datasets.

TECHNICAL SKILLS

Programming: Python (JAX / Flax, PyTorch, Hugging Face), Bash, LaTeX
World models and inference: extending DreamerV3 / RSSM in JAX; SMC with EIG-based experiment selection
Compute and tooling: SGE; multi-GPU LoRA training with HF Accelerate (3×A100); Docker / Apptainer; Git; Linux